

IntelliTwin: An AI-Driven Digital Twin Framework for Predictive Maintenance of Engineering Assets

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Abstract—Industrial equipment failures cause substantial operational disruption, production losses, and emergency maintenance costs. Reactive maintenance acts only after failure; fixed-interval preventive maintenance may service equipment regardless of actual condition. This paper proposes IntelliTwin, an AI-driven Digital Twin framework for predictive maintenance of engineering assets. The framework combines IoT sensor measurements (vibration, temperature, motor current) with a continuously updated virtual asset representation. Machine learning provides anomaly detection, fault identification, degradation estimation, and Remaining Useful Life (RUL) prediction. The resulting health information is converted into risk scores, maintenance alerts, and decision-support outputs. The architecture spans sensor acquisition, data processing, Digital Twin synchronisation, AI/ML inference, visualisation, and decision-support layers, supporting a transition from reactive toward condition-based and predictive maintenance.

Index Terms—Digital Twin, Predictive Maintenance, Machine Learning, Artificial Intelligence, Industrial IoT, Remaining Useful Life, Anomaly Detection, Condition Monitoring.

I. INTRODUCTION

Engineering assets such as pumps, motors, turbines, compressors, and conveyor systems form an important part of industrial operations. Their reliability and availability directly influence production output, service quality, and operational efficiency. Unexpected equipment failures can result in production stoppages, safety risks, emergency repairs, and damage to associated components.

Traditional maintenance strategies are based on reactive maintenance or scheduled preventive maintenance. Reactive maintenance waits until an asset fails, while scheduled maintenance services equipment at predetermined intervals. Although useful, these approaches do not fully exploit the continuous condition information available from modern sensors [1], [2].

The development of Industrial Internet of Things (IIoT) technologies has enabled continuous collection of vibration, temperature, current, pressure, and other operational measurements. Simultaneously, artificial intelligence and machine learning can learn patterns from sensor data, identify abnormal behaviour, estimate degradation, and support failure prediction.

A Digital Twin provides a dynamic virtual representation of a physical asset synchronised with real-world data [3]. When combined with AI/ML, the Digital Twin can represent the current health state of equipment and support forward-looking predictions. IntelliTwin is proposed as an integrated framework connecting sensor data, Digital Twin modelling, AI-based health analysis, and maintenance decision support.

The proposed work aims to develop a system that continuously monitors an engineering asset, detects early deviations from normal operation, estimates failure risk and RUL, and provides actionable maintenance information. The project is at the proposal/design stage; implementation and experimental validation are left for subsequent work.

II. LITERATURE REVIEW

A. Digital Twin Technology

Digital Twin technology represents a physical asset through a connected virtual model that is continuously updated using operational data. Fuller et al. [4] provided a comprehensive overview of Digital Twin enabling technologies, challenges, and open research, characterising the concept as the effortless bidirectional integration of data between a physical and virtual machine. Tao et al. [2] surveyed the state-of-the-art of Digital Twins in industry, highlighting lifecycle management and maintenance decision support applications. Kritzing et al. [5] offered a categorical literature review and classification differentiating digital models, digital shadows, and full Digital Twins. A typical architecture includes the physical asset and sensors, the digital representation, and the communication/data layer connecting them.

B. Predictive Maintenance

Maintenance approaches range from reactive to condition-based and predictive maintenance. Carvalho et al. [6] conducted a systematic review of machine learning methods for predictive maintenance across sensor types and asset classes. Rathore et al. [7] reviewed the role of AI, machine learning, and big data in digital twinning, demonstrating how integrated approaches enable early fault identification and reduce unnecessary preventive interventions. De Benedetti et al. [8] proposed an EDA-mapped sensor framework for anomaly detection showing timely alert generation across multiple monitored machines.

C. AI/ML for Asset Health Monitoring

Machine learning extends Digital Twin capabilities by learning normal operating behaviour and detecting deviations in multivariate sensor data. Roy et al. [9] demonstrated the

effectiveness of autocorrelation-aided random forest classifiers for bearing fault detection from vibration signals. Nassif et al. [10] provided a systematic review of ML-based anomaly detection, covering isolation forest, one-class SVM, autoencoders, and deep learning approaches applicable to IIoT environments. For RUL estimation, Ma and Mao [11] proposed a deep-convolution-based LSTM network achieving state-of-the-art performance on the CMAPSS benchmark. Yang et al. [12] addressed uncertainty quantification in LSTM-based bearing RUL prediction. Zhou et al. [13] introduced a GRU network with reinforced memory for improved long-horizon RUL prediction. Wu et al. [14] demonstrated that LSTM with Bayesian and Gaussian processing is effective for anomaly detection in industrial IIoT sensor streams.

D. Research Gap

Existing Digital Twin and predictive-maintenance approaches demonstrate the value of combining physical asset information, sensor data, and intelligent analytics. Liu et al. [15] identified the lack of modular, end-to-end integration frameworks as a key gap. IntelliTwin is proposed to address this integration through a modular pipeline architecture connecting all layers from sensor acquisition to maintenance decision support.

III. PROPOSED METHODOLOGY

IntelliTwin follows a pipeline-based architecture that connects the physical asset to maintenance decision support. The planned data flow is:

Physical Asset → *IoT Sensors* → *Data Acquisition* → *Preprocessing* → *Digital Twin + AI/ML Engine* → *Health/Fault Prediction* → *Dashboard* → *Maintenance Decision*.

A. Physical Asset and Sensor Layer

The system monitors a centrifugal pump and motor as the primary case study. Sensors provide vibration, temperature, and motor current measurements. Pressure and operating-time data may also be integrated where available, providing the condition information required for continuous health monitoring.

B. Data Acquisition

Sensor readings are collected through an acquisition layer and time-stamped to align measurements from different sensors. The design supports industrial communication protocols including MQTT, OPC-UA, and MODBUS TCP, depending on the hardware and experimental setup.

C. Data Preprocessing

Raw sensor data is cleaned prior to analytical processing. Steps include noise filtering, missing-value handling, outlier treatment, resampling, normalisation, and timestamp alignment. Quality flags are retained to distinguish reliable observations from uncertain data in downstream models.

D. Feature Extraction

Processed signals are transformed into informative features. For vibration data, candidate features include Root Mean Square (RMS), kurtosis, peak energy, and frequency-domain characteristics via Fast Fourier Transform (FFT). Temperature trends and rolling-window statistics (mean, variance, skewness, peak-to-peak amplitude) are also computed.

E. Digital Twin Model

A virtual representation of the asset is created using equipment parameters, component specifications, and maintenance history. Following the asset-state synchronisation paradigm described in [2] and [4], the Digital Twin maintains variables such as vibration level, temperature, load, speed, and estimated health condition, updating as new sensor observations arrive.

F. AI/ML Engine

The AI/ML layer analyses extracted features and Digital Twin state variables. Anomaly detection methods [10], [14] identify deviations from normal behaviour. Classification models—random forest [9] and gradient boosting—are used for fault identification. LSTM [11], [12] and GRU [13] models are evaluated for RUL estimation. Final model selection depends on dataset characteristics and cross-validation results.

G. Health Estimation and Decision Support

Model outputs are combined into a composite health indicator, failure-risk score, and estimated RUL. A maintenance dashboard presents the current asset condition, detected anomalies, probable fault categories, and recommended next actions. Alert thresholds and escalation logic will be finalised after implementation and experimental validation.

H. Model Training and Validation

Sensor data will be partitioned into training and test sets using time-based or asset-based splits to prevent data leakage. RUL models will be evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Classification models will be assessed using precision, recall, and F1-score. Early-warning lead time is a key operational metric. Numerical performance results will only be reported after full implementation.

IV. CONCLUSION

This paper has presented IntelliTwin, an AI-driven Digital Twin framework for predictive maintenance of engineering assets. The proposed architecture integrates IoT sensor acquisition, data preprocessing, feature extraction, Digital Twin state synchronisation, and an AI/ML inference engine within a single modular pipeline. The framework supports condition-based and predictive maintenance by providing real-time health indicators, fault classifications, and Remaining Useful Life estimates. Future work will focus on implementation with a centrifugal pump and motor test-bed, followed by experimental validation and comparison with

baseline maintenance strategies.

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